

Project Information

- **Group size:** up to **3 students** per group
- Your group can **select one of three given projects** and work on it:
 1. Insurance Problem
 2. Coffee Problem
 3. Restaurant Problem

Submission Guidelines

- Only **one submission per group** is required.
- Clearly **list all group members' names** in the submission.
- **Group size:** up to **3 students**.
- Submit a **Jupyter Notebook file (.ipynb)** containing your full code and analysis on Blackboard
- **Deadline:** May 4, 11:59 PM.

Grading Rubric

1. Exploratory Data Analysis (15 points)

Provide a thorough descriptive analysis of the dataset. This may include:

- Descriptive statistics such as mean, median, correlations etc.
- Scatter plots, box plots, bar charts
- Clustering or other exploratory techniques
- Any additional relevant analysis

2. Modeling (15 points)

- Develop at least one **final model**.
- Compare its performance with **alternative models**.

3. Model Interpretation and Performance (15 points)

- Interpret the results of your model clearly.
- Evaluate model performance using appropriate metrics.
- Justify your evaluation approach.

4. Key Findings (15 points)

- Clearly summarize your insights.
- Address the main questions or objectives of the analysis.

5. **Code Evaluation (20 points)**

- Notebook should be **well-organized, readable, and error-free**.
- Include **sufficient comments/documentation**.
- Ensure all code needed to reproduce results (plots, outputs) is included.

6. **Coding Practices (20 points)**

- Code should be **efficient and well-structured**.
- Use **PySpark functions wherever appropriate**.
- Limited use of Pandas or other libraries is acceptable when PySpark alternatives are not suitable.